
What the Final Patch Hides: Event-Level Regression Measurement for Coding-Agent Harnesses

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Abstract

1 We instrumented an unmodified coding agent to replay the benchmark’s previously-
2 passing tests after every edit, and found that transient regressions are common but
3 almost never survive to the graded patch — while replacing our own harness’s
4 model-written check with an execution-grounded one recovers the resolve that
5 check was destroying (13 to 26 of 40; exploratory, and circular by construction).
6 Coding agents are graded on the final patch alone; the instrument commits the
7 working tree after every mutating tool call and replays the benchmark’s oracle at
8 every commit inside its pinned container. We applied it to mini-swe-agent 2.4.6,
9 unmodified: its own loop, prompts, model layer, budgets, and submission protocol,
10 with only its environment object replaced. On 40 SWE-bench Verified tasks it
11 resolves 34 of 40 with `claude-sonnet-4-6` and 33 of 40 with `gpt-5.6-sol`,
12 ends all 80 runs with a changed tree, and breaks previously-passing test functions
13 in 11 of them — 135 counted per run, 100 distinct — of which exactly one is still
14 failing when the official grader reads the final tree. Its own testing does not close
15 the gap: over the window in which each test was broken, no test command ran the
16 broken test in 6 of the 10 classifiable cases, and in the other 4 one ran it and printed
17 the failure by name — so a method reading only the agent’s logs is structurally
18 blind to most of them. It holds in our own harness under rollback (146 test functions
19 broken across 8 of 47 runs, one still failing) and on SWE-bench Live, with two
20 caveats we make measurable: rollback leaves clean trees partly because 17 of its
21 37 runs ship a tree they never changed, and a model-stack difference on Verified
22 does not survive a change of scaffold. The blindness runs both ways — nine of the
23 ten trees that killed the suite recorded no timeline event. Gating each step on the
24 repository’s own tests instead resolves 65% with no contaminated final tree, and
25 70% when the gate reads only half of those tests, with no failure among the 2,585
26 previously-passing ids it never read. We will release the instrument, its validation
27 protocol, and the archived timelines; Appendix A catalogues 28 measurement
28 defects met while building it, 23 of which produced plausible numbers rather than
29 errors.

30 1 Introduction

31 Benchmarks for coding agents grade the final patch: SWE-bench [1] and its Verified subset [2] apply
32 the submitted patch and report whether the target tests pass and the previously passing ones still
33 pass; work on agent-caused regressions counts what the submitted patch breaks [8]. That convention
34 cannot see what happens while the agent works — an agent that breaks a test at step three and repairs

35 it at step five submits a clean patch — so two questions have no answer: how often do agents break
36 existing behaviour during a run, and how much survives?

37 The answer is less than we expected and not where we expected: agents break the oracle often and
38 almost always repair it, so what the endpoint misses is transient rather than shipped. We measure the
39 timeline instead of the endpoint — every mutating tool call becomes a commit on a git reference the
40 agent cannot see, and the oracle is re-run at every commit inside the benchmark’s pinned container.
41 This is the observation primitive an agent operating system needs before it can build a recovery
42 primitive. Our contributions are:

- 43 1. **An instrument for event-level regression measurement**, applicable to an agent we did
44 not write: every mutating tool call becomes a commit on a hidden git reference, and the
45 benchmark’s previously-passing oracle is replayed at every recorded state inside its pinned
46 container (Section 3).
- 47 2. **The measurement on an unmodified public scaffold**. mini-swe-agent 2.4.6 keeps its own
48 loop, prompts, model layer, budgets, and submission protocol; only its environment object
49 is ours. It resolves 34 and 33 of 40 under two model families and breaks 135 oracle test
50 functions counted per run (100 distinct) across 11 of the 80 runs, one still failing at grade
51 time. In 6 of the 10 classifiable cases its own test commands never ran the test it broke
52 (Section 5.1).
- 53 3. **The same measurement across recovery policies**. The undercount holds in our harness and
54 on SWE-bench Live, but nearly half the rollback arm’s runs ship a tree they never changed,
55 so a clean final tree is partly no tree at all (Sections 5.2, 5.4, 5.6).
- 56 4. **Recovery policy is downstream of monitor precision**. With a model-written check,
57 rollback is strictly harmful (13 against repair-in-place’s 24); replacing it with the repository’s
58 own tests recovers the loss (26, no contaminated tree) at about 30 seconds per run —
59 exploratory, and circular by construction (Section 5.5).
- 60 5. **A catalogue of 28 measurement defects** found while building the instrument (Appendix
61 A); 23 produced a plausible number rather than an error.

62 All measurements are exploratory, on development slices excluded from future confirmatory study.

63 2 Background and related work

64 **Benchmarks and their validity**. SWE-bench [1] grades a patch by tests that must go from failing to
65 passing and tests that must keep passing; Verified [2] is the human-screened subset, deprecated as
66 a capability metric [46]. Concerns about it have accumulated — contamination and memorisation
67 [6], plausible-but-wrong patches overstating resolve by 6.2 points [37], and insufficient tests, which
68 UTBoost [7] augmented to expose mislabelled patches affecting 24.4% of leaderboard entries —
69 with rolling [3, 5], private [4], and long-horizon [23] alternatives in response. TDAD [8] measured
70 regressions in *submitted* patches on Verified — 6.1% of previously-passing tests, 30.2% of non-
71 empty patches — and reduced them with an impact-analysis map. Process-level benchmarks score
72 intermediate states from logs [35, 36], AgentLens [38] naming regression cycles in 10.7% of passing
73 OpenHands runs — close to our 11 of 80, but a log-derived rate is bounded below by what the agent
74 chose to run, and in 6 of our 10 classifiable cases it ran nothing that would have executed the broken
75 test. We execute the oracle at every mutating call instead.

76 **Agent scaffolds and recovery**. SWE-agent [9], OpenHands [10], Agentless [11], and mini-swe-agent
77 [12] span the scaffold design space [13, 14]. Within-episode recovery appears as self-reflection [15,
78 16], progress-gated recovery [17], checkpoint repair [18], and aligned checkpoints [41]; provenance-
79 based recovery is surveyed in [19]. Closest to us, Kim et al. [39] match logged edits against the
80 gold diff and recover agents that reach a gold-identical patch and then destroy it; Gao et al. [40]
81 bind verifier evidence to code states; neither counts regressions or compares policies. Running
82 the repository’s tests during a run is deployed practice [43, 44]. Operating-system framings [20]
83 and the branch-context primitive [42] supply scheduling and fork, commit, and abort, but do not
84 observe the tree between actions. The regression problem is program repair’s plausible-versus-correct
85 patch problem [26, 28, 29], where regression tests are the main defence against overfitting [27];
86 trajectory-level evaluation [21, 22] argues resolve rate alone is uninformative. Public leaderboard
87 trajectories record actions but not the tree at each step; the instrument commits the tree.

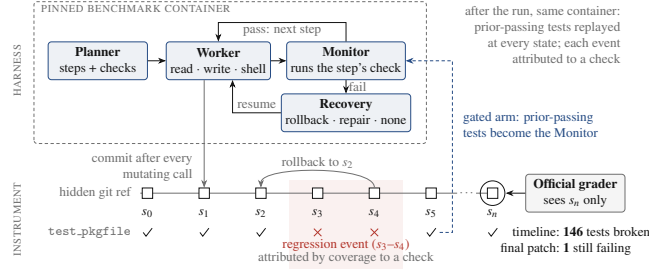


Figure 1: The harness and the instrument. Every mutating tool call commits the tree to a hidden git reference; after the run, the instance’s previously-passing tests are replayed at every state. The test row shows one regressing at s_3-s_4 and repaired by a rollback to s_2 , which the official grader, reading only s_n , never sees; the counts are the GPT-5.6 rollback arm’s. In the gated arm (dashed), the timeline’s tests serve as the monitor.

3 Instrument

Figure 1 shows the harness and the instrument. **Observational timeline.** After every mutating tool call the working tree is committed to a git reference outside the agent’s view (a private index keeps its own git status unchanged); rollbacks and the run’s end are observations too, and an archived run can be re-measured without re-running the agent. The agent’s tools and the harness’s checks execute inside the same pinned container as the replay, synchronised to the host tree the timeline records (Appendix A.2).

Exhaustive replay. For each observation the instance’s previously-passing (PASS_TO_PASS) tests run inside the pinned container against that observation’s tree; a regression event is a test that passes at one observation and fails at a later one. Every observation is replayed rather than bisected, because a recovery policy makes verdicts non-monotone. Replay is scoped to the files holding those tests, which keeps it affordable on Verified (about 26 CPU-minutes per sweep); Live’s larger oracles cost tens of minutes per instance. Infrastructure failures during replay, and tests already failing in the unmodified image, are recorded as missing observations rather than failures.

Validation. Five checks run before any paid experiment (negative and positive controls, a flake screen, an unknown-rate ceiling, a baseline liveness check), plus a golden check that drives the gold patch through the agent’s real tool path and a re-scoring control that re-measures an archived timeline with no model calls; re-scoring four arms’ bearing runs reproduced their episode counts and probe sets exactly, and supplies the matched-unit columns of Table 1. Appendix C gives the detail.

4 Experimental setup

Benchmark and slice. SWE-bench Verified is a measurement substrate here, not a leaderboard: its pinned images and oracles make the measurement possible. A 40-instance development slice (16 from django), stratified and fixed before any measurement, was used throughout and is excluded from future confirmatory study; the GPT-5.6 stack also ran its first 10 as a calibration, and plain rollback ran twice. SWE-bench Live [3], built monthly from live repositories, is the second substrate, graded by its own harness whose rules we mirror; its slice was fixed before any outcome (Appendix B) and its oracles are about twenty times larger (median 1,189 against 58 tests).

Harness. A planner decomposes the task into steps with verification commands; a worker executes each with three tools (read, write, shell); a monitor runs the verification, and on failure the recovery policy acts: **rollback** (reset to the last verified checkpoint and retry), **repair-in-place** (retry from the failed tree), or **no recovery** (the monitor’s FAIL is final and the run ends). Runs have a \$4 work-cost cap. One instance under one policy is a cell; each (model stack, policy, substrate) triple is an arm.

Public scaffold. mini-swe-agent 2.4.6 [12] runs unmodified — its own loop, prompts, model layer, 250-step and \$3 billed-cost caps, and submission protocol. Only its environment object is replaced, so every bash command executes in the benchmark’s pinned container and the tree is committed to the hidden reference after each command. The seam is inert rather than invisible: the agent’s git

124 `diff` and `git status` show its own uncommitted edits and none of the instrument’s commits, but
125 the synthetic baseline carries a marker name many runs saw (Appendix C). Six departures from its
126 leaderboard configuration are listed there; we claim no equivalence to its published resolve rate, and
127 because its budget differs in kind from our cap, no comparison between harnesses is tested. A grid
128 census (Appendix C) puts its sampling inside our arms’ range — a median of 3 and 2 tree-changing
129 observations per run against our 2 to 5 — while it issues 9 to 15 bash commands per observation,
130 most of them reads.

131 **Models.** Two frontier stacks under identical harness, policy, instances, and caps: `claude-opus-4-7`
132 with `claude-sonnet-4-6`, and `gpt-5.6-sol` with `gpt-5.6-terra` (planner, worker), recorded in
133 every run manifest.

134 **Outcomes.** Resolve is the official grader’s verdict on the final patch. Events are reported at three
135 levels because a few runs produce most: incidents (observations at which at least one oracle test
136 broke, the co-primary endpoint), declared events (one per test function and onset), and bearing runs
137 (runs with at least one event). Because an event and a final-state failure are different objects, the
138 undercount is stated in matched units: test functions broken during a run against those still failing at
139 the end. Contamination counts oracle tests failing in the graded patch, net of baseline-dead; a tree
140 that kills the suite is graded as failing every test but is a missing observation for event counting, so a
141 contaminated cell need not be a bearing run.

142 **Pre-declaration.** The event unit, the proceed criteria, and the recovery-policy contrast were declared
143 before the relevant data existed, and the analysis was committed before either comparison arm
144 finished; one amendment preceded unblinding and one grading correction followed it (Section 5.4,
145 Appendix B).

146 5 Results

147 5.1 An unmodified public scaffold breaks tests and puts them back

148 `mini-swe-agent 2.4.6` ran unmodified on the same 40 instances: with `claude-sonnet-4-6`, 34 of 40
149 resolved (35 rot-aware) over 2,101 bash commands for \$31.38, four runs ending at its own \$3 cap —
150 three of those still grade resolved, since resolve reads the final tree, not the patch the scaffold never
151 submitted; with `gpt-5.6-sol`, 33 of 40 (34) over 937 commands for \$12.13. No run of the 80 ended
152 byte-identical to its start.

153 The timeline recorded 56 events in 5 of the 40 Claude runs and 79 in 6 of the 40 GPT-5.6 runs, on 8
154 instances, 3 bearing under both. Those 11 runs broke 135 oracle test functions counted per run (100
155 distinct, the stacks overlapping on the three shared instances), and when the grader read the final tree
156 exactly one was still failing: `test_pkgfile` in `pytest-6197` (Claude 0 of 56; GPT-5.6 1 of 79). The
157 scaffold has no monitor, no checkpoint, and no rollback, so every one of those closures is an ordinary
158 forward edit the agent chose to make. They are short and concentrated: the eleven stayed open for 1
159 to 22 bash commands (median 4), and three runs supply 81% of the 135. Much of what the endpoint
160 misses is therefore a transient inside a single logical edit — but a transient is not free: 14% of what
161 these eleven runs cost (\$1.29 of \$9.52) was spent while a regression was open.

162 **Would its own testing have caught it?** For each breaking edit we read the scaffold’s trajectory
163 over the window the test was broken — the breaking command up to, but not including, the repair
164 — asking whether a test runner ran, whether any run would have executed *this* test, and whether its
165 output showed the failure (Appendix C; coverage is three-valued, *unknown* for runs that stopped early
166 or were truncated). In 6 of the 10 classifiable cases the agent’s own test commands never ran the
167 broken test; in the other 4 they ran it and printed the failure by name, and it was repaired anyway (the
168 eleventh is unknown, a `pytest -x` run that stopped earlier). Self-testing is blind to the first group
169 and ignored on the second; the endpoint sees neither. With eleven cases this is a case series, not a
170 rate.

171 5.2 The same undercount in our harness, and what the clean tree costs

172 Under rollback the Claude stack resolved 13 of 40 (32.5%, exact 95% CI 18.6% to 49.1%) for \$34.24
173 and the GPT-5.6 stack 13 of 40 for \$8.14; a circuit breaker stopped three cells and a synchronisation
174 defect lost two, so 37 were attempted and 35 graded. One instance is unresolvable under the current

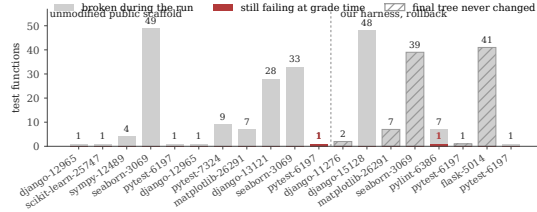


Figure 2: Every run that broke a previously-passing test, in matched units: test functions broken on the timeline against those still failing when the grader reads the final tree (dark). Left, the unmodified public scaffold under both models; right, our rollback arm, where hatched bars are runs whose final tree is byte-identical to their start.

175 parser (Appendix A.3), capping every arm at 39, the scaffold included. The median rollback run
 176 makes 4 mutating tool calls under GPT-5.6 and 3 under Claude.

177 Eight of the 47 GPT-5.6 rollback runs (the sweep and a 10-instance calibration) broke a previously-
 178 passing test: 146 test functions counting each run separately (145 distinct), of which one —
 179 `test_can_read_toml_env_variable` in `pylint-6386` — was still failing at grade time. Three
 180 storms supply 88% of the 184 declared events over all 47 runs; Table 1’s 140 is the 37-run sweep
 181 alone. Declared events are reported as exposure only. Figure 2 shows every bearing run on both sides.

182 Not every clean tree is work. Excluding the hidden reference, 17 of the 37 rollback runs end
 183 byte-identical to their start, as do 19 of the Claude stack’s 40, 4 of 40 under the gate, and 0 under
 184 repair-in-place. Four of the six bearing sweep runs shipped nothing; over the two that delivered a
 185 patch, 1 of 55 broken test functions is still failing — which is why the public-scaffold arm, none of
 186 whose runs ends unchanged, carries the claim.

187 The blindness runs both ways. A break that stops the suite from being collected opens no timeline
 188 event, because a test the parser never mentions is a missing observation rather than a failure, while
 189 official grading scores such a patch as failing everything. Nine of the ten suite-killed trees across all
 190 arms recorded no event: the two instruments have complementary blind spots.

191 5.3 On Verified, a model-stack difference that does not generalise

192 On the same 40 instances, under the same harness and policy, the Claude stack produced no regression
 193 events in 40 runs, the GPT-5.6 stack 140 in 6 of 37 (one-sided Fisher $p = 0.010$, a single pairwise test
 194 on six bearing runs — reassign three of them and $p = 0.11$). Two controls bound it: exposure differs
 195 about twofold, not 140-fold (147 opportunity observations against 218), and the zero is measured,
 196 not a detection failure, since re-scoring that stack’s canary reproduces its three events (a replicate
 197 sweep is in Appendix C). It does not generalise — the same `claude-sonnet-4-6` worker under
 198 unmodified mini-swe-agent is bearing on 5 of 40 with 56 events (Section 5.1), and it is not detected
 199 on Live — so we report a measurement on one slice, not a property of either model; model and
 200 adapter are confounded.

201 5.4 Recovery policy: what the clean final tree costs

202 On the primary endpoint, contamination paired by instance, no recovery was worse than rollback
 203 on 9 instances and better on 1 (exact sign test $p = 0.021$); repair-in-place was worse on 5, better on
 204 1 ($p = 0.22$). Incident exposure, the co-primary, did not differ ($p = 0.45$ and 1.0). The arms cannot
 205 show the policy *caused* the difference: no recovery ends at the monitor’s first FAIL, so its regressions
 206 arrive on the last tree-changing observation with nothing after them, and its 19-of-19 persistence is
 207 the stopping rule as much as the policy.

208 Rollback pays for its clean tree in delivered work: 13 of 40 against 24 and 22, losing nine discordant
 209 pairs to repair-in-place and winning one (McNemar $p = 0.021$). The loss is the monitor’s — of the 18
 210 failed rollback runs, 15 failed at the first step, and 9 of those instances were resolved by an arm that
 211 kept work the check rejected and the grader accepted — and 17 of 37 runs ended unchanged.

Table 1: Every arm on both substrates. **empty**: runs ending byte-identical to their start. **events**: declared (test function, onset) pairs — exposure only, never a denominator. **broken** → **left**: over the bearing runs, oracle tests broken during the run against those still failing at grade time. **contam.**: graded cells failing an oracle test, net of baseline-dead. The public-scaffold rows ran under mini-swe-agent’s own budget, not our cap, so no comparison between harnesses is tested. Definitions, incidents, per-row composition and provenance: Appendix C.

substrate	arm	resolved	empty	events	bearing	broken → left	contam.	spend
Verified	mini-swe-agent, Claude	34	0/40	56	5/40	56 → 0	0	\$31.38
Verified	mini-swe-agent, GPT-5.6	33	0/40	79	6/40	79 → 1	1	\$12.13
Verified	GPT-5.6 rollback	13	17/37	140	6/37	104 → 1	1	\$8.14
Verified	Claude rollback	13	19/40	0	0/40	—	1	\$34.24
Verified	GPT-5.6 gated	26	4/40	70	5/40	62 → 0	0	\$10.51
Verified	GPT-5.6 split-gated	28	3/40	14	3/40	14 → 0	0	\$9.40
Verified	GPT-5.6 repair-in-place	24	0/40	84	7/40	83 → 4	5	\$14.73
Verified	GPT-5.6 no recovery	22	1/40	19	3/40	19 → 19	9	\$4.40
Live	GPT-5.6 rollback	0 (2)	25/39	156	6/39	147 → 0	0	\$10.29
Live	Claude rollback	2 (3)	24/39	49	3/39	48 → 0	1	\$49.80
Live	GPT-5.6 gated	1 (4)	9/39	280	3/39	270 → 0	2	\$11.49
Live	GPT-5.6 no recovery	4 (6)	3/39	30	4/39	30 → 30	4	\$8.95

212 **Disclosure.** The first unblinding gave $p = 0.375$, because seven cells whose final trees killed the suite
213 had been dropped as ungradable; correcting the grader and re-grading those archived trees gives the
214 0.021 above — a rule change made after the data were seen (Appendix B).

215 5.5 On Verified, regression-gated rollback removes the tradeoff

216 A fourth arm, added after unblinding (exploratory), replaces the planner-written check with the
217 repository’s own oracle, run after every attempt: a step is rejected only if a test that passed at the
218 start now fails. It resolved **26 of 40 (65.0%)**, above rollback’s 13 and the ungated arms’ 24 and 22,
219 below the 34 and 33 mini-swe-agent reaches under a budget differing in kind (Table 1), with **no**
220 **contaminated final tree** and 4 empty patches against rollback’s 17. The guarantee is cheap: at 4 to 8
221 seconds per observation and a median of 4 per run, about 30 seconds of wall-clock and \$2.37 more
222 model spend than plain rollback. Against rollback on 35 shared instances it won 10 discordant pairs
223 and lost 1 ($p = 0.012$); against repair-in-place it resolves 26 to 24 and is better on contamination 5 to
224 0. What the gate offers is a guarantee rather than a rate, and it is circular: its tests are the benchmark’s
225 own oracle (median ratio 1.00), so a clean tree follows whenever the gate saw the final tree and the
226 grader reads what the gate read. A fifth arm weakens that: the gate reads a deterministic half of the
227 ids (2,278) and never the other half (2,585), on which the grader also rules [45]. The split is over ids,
228 not files, so a break reaching a held-out id usually reaches a watched one; the held-out half bounds
229 the leak rather than sampling it, and a file-level or non-oracle gate would settle it. It resolved **28 of 40**
230 **(70.0%)**, indistinguishable from the full gate ($p = 0.73$) and better than rollback (13 to 2, $p = 0.007$),
231 with no held-out failure apart from the parser-capped instance.

232 5.6 Replication on SWE-bench Live

233 Live keeps the undercount and loses the resolve rates: at most 6 of 40 even rot-aware, so its recovery
234 contrast is uninformative, and 25 of 39 rollback runs shipped an unchanged tree. The counts scale
235 with the oracle — Live’s is twenty times Verified’s, and rollback broke 147 test functions across 6 of
236 39 runs against 104 across 6 — none still failing, while no recovery left all 30 of its own ($p = 0.125$).
237 The stack difference did not replicate (3 of 39 against 6, $p = 0.24$).

238 6 Discussion

239 **Why timeline regressions matter.** A repaired regression is not free: 27% of what the eight Verified
240 bearing runs cost (\$0.63 of \$2.33), and 14% of the scaffold’s, went to work done while one was
241 open, and an endpoint-only benchmark cannot tell a run that broke nothing from one that broke 49
242 tests and repaired them. The events are the step-level ground truth patch verifiers [24] and process
243 reward models [25] need, and they tell a recovery primitive [42] when to abort. Two rules would have
244 prevented most of Appendix A’s defects: record infrastructure failures as missing observations, and
245 require positive evidence that the measurement path is live. **Limitations.** Every rate is an estimate on
246 a 40-instance slice with one sweep per arm; event counts are lower bounds both ways (Appendix C).

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338 A The measurement defect catalogue

339 A.1 The catalogue by mechanism

340 Each row is a defect found while building the instrument. **S**: it produced a plausible number rather
341 than an error. **G**: it was present while the test suite passed (A4 is the suite). **H**: it was findable only
342 on a clean host. Class A rows depend on the machine the code runs on; class B rows render an
343 infrastructure failure as a measurement; class C rows measure something other than the event as
344 defined; class D rows consume a producer’s output without checking the producer.

#	Defect	Consequence	S	G	H
A1	Shadow commits inherited the machine’s git identity	timeline silently empty on any clean machine	✓	✓	✓
A2	Validation probe invoked bare python	every probe exit 127 on a clean host	✗	✓	✓
A3	Unpinned SDK resolved a different major version	two machines ran different code	✗	✓	✓
A4	Test fixtures verified with bare pytest from PATH	15 tests fail on a clean host as harness defects	✗	—	✓
A5	A benchmark scorer invoked bare python	score 0.0 from a missing interpreter	✓	✓	✓
A6	Agent executed on the host; measurement in the container	26 of 28 zero-step runs; see A.2	✓	✓	✓
B1	Output marker used shell :	a 13/13 run scored as 13 errors	✓	✓	
B2	Results on stderr, markers on stdout	one framework’s results outside the parsed slice	✓	✓	
B3	Probe diffed against a deleted commit	every graded test a hole	✓	✓	
B4	Interrupted mirror clone accepted with zero commits	later instances fail far from the cause	✓	✓	
B5	Container workdir unvalidated	every exec exit 127	✓	✓	
B6	Isolation removed loopback, not only the internet	23.5% of the oracle dead	✓	✓	
345 B7	Provider cache served removed containers	later cells replay against nothing and score clean	✓	✓	
B8	Tree reset reverted image build-time edits	one repository family’s oracle dead	✓	✓	
C1	Bisection assumed monotone verdicts	recovered regressions invisible	✓	✓	
C2	Declared event unit not implemented	rate off by up to 2.7×	✓	✓	
C3	Rollbacks produced no observation	recoveries invisible to the timeline	✓	✓	
C4	"Persists past the step boundary" undefined	one reading excludes the events of interest	✓	✓	
D1	Audit field never populated	0 tool errors reported, always	✓	✓	
D2	Malformed planner output crashed the run	33% of runs lost as task failures	✗	✓	
D3	Dependency install counted as agent work	attribution vacuous	✓	✓	
D4	Join took the first observation, not the last	failures pinned to the wrong tree	✓	✓	
D5	Executed config rebuilt from a name	manifest described a different run	✓	✓	
D6	Git handles never released	sweeps die after ~400 cells	✓	✓	
D7	Console re-walked the tree per run	presented as a dead server	✗	✓	
D8	Detection events lacked the ids attribution joined on	attributed detection always unknown	✓	✓	
D9	Absent coverage rendered as measured silence	unmeasured regressions reported as silent	✓	✓	
D10	Output-capped turns executed	half-written file; a 78-event storm	✓	✓	

346 Totals: 23 of 28 silent; 27 of 28 present under a passing suite; 6 of 28 findable only on a clean host.
347 Ten further defects found after this census closed are described where they arose (Sections 5.2 and
348 5.4, Appendices A.3 and C); the rest will be released with the code.

349 A.2 The defect that invalidated a conclusion

350 After nineteen fixes, every validation check passed, the re-scoring control reproduced archived
351 episode counts exactly, and three pilots (80 runs) measured an event rate far below the pre-declared
352 proceed criterion. We drafted the conclusion the rule directed: the benchmark offered too little
353 opportunity, so switch benchmarks. The conclusion was wrong. The harness ran the agent on the host
354 in a bare source checkout, while every probe and every gate ran inside the pinned container. On the
355 host, `import matplotlib` in that checkout succeeds by importing the uncompiled source tree as a
356 namespace package, and everything downstream fails in ways indistinguishable from an incompetent

agent. Twenty-six of 28 zero-step runs traced to this. The checks had validated the measurement path and never the agent’s execution path. The fix was to route the agent’s tools and checks through the same container as the replay, and to add a per-cell environment parity check that runs before any model call.

361 A.3 Mechanisms in public infrastructure

362 Class A: OpenHands issues #4235 and #7044 [31], environment construction failures against SWE-
363 bench images. Class B: UTBoost’s finding that the held-out oracle is insufficient at leaderboard scale
364 [7], and, on SWE-bench Live, baseline tests that fail in the unmodified image because the calendar
365 passed a deprecation date encoded in the instance. Class C: final-state regression measurement [8].
366 Class D: the SWE-bench grader accepting forged test output on stdout [30], and, at the harness’s
367 current revision, a parser that keys parametrised ids by their full text while the dataset stores them
368 truncated at the first space (the earlier parser’s behaviour), so 16 previously-passing ids of one instance
369 in our slice match no output and every submission to it grades as unresolved.

370 B Pre-declaration and analysis

371 The event unit is one (test function, onset observation) pair with parametrised variants collapsed. The
372 gates for proceeding on a substrate were an event rate of at least 0.30 per run and a bearing fraction
373 of at least 25%. Both stacks failed the conjunction (bearing 0% and 16.2%); the gates were then
374 re-declared per (substrate, model stack) before the contrast was unblinded, and no arm reported here
375 is confirmatory. The contrast’s primary endpoint is final-state contamination, paired by instance, exact
376 two-sided sign test; the co-primary is incident exposure. The analysis script was committed before
377 either comparison arm finished. The regression-gated arm was added after the contrast was unblinded
378 and is exploratory. No multiplicity adjustment was planned or applied. Budget exhaustion is scored
379 as failure. Instances whose baseline oracle records failures in the unmodified image are excluded
380 from resolve-rate comparability claims, and their dead tests are typed as missing observations for
381 event counting.

382 **Compute.** Every run, replay, and grade executed on one 12-vCPU, 22 GB x86 virtual machine
383 running Docker, with no GPUs; the six harness arms on Verified in Table 1 cost \$81.42 in model spend
384 and the two public-scaffold arms \$43.51, plus \$6.87 for the second rollback sweep, the calibration
385 and pilots a comparable amount, and the Live sweeps are reported with their own spend in Table 1.

386 **Slice construction on SWE-bench Live.** Fixed before any outcome: pytest-parsed instances with
387 27 to 3,000 previously-passing tests, at most four per repository, earliest first; the first 40 of 48
388 candidates whose image passed a model-free environment check. No coverage maps are built for
389 Live, so attribution there is unknown by construction.

390 **Attribution (Verified only).** A detected regression is classified three ways: *attributed* if a failing
391 harness check and the broken test both exercise a file the agent changed at that observation (per-
392 instance coverage map at the base commit), *co-occurring* if some check failed while the regression
393 was open without such a link, and *unknown* if the map cannot say. Of the calibration’s 57 raw
394 episodes, 39 were attributed to a failing check, 16 co-occurred with one, and 2 were silent; 48 of the
395 sweep’s 162 had none. The public-scaffold arm has no monitor, so attribution is undefined for it and
396 no silence statistic is reported.

397 **Grading correction between unblindings.** The first unblinding of the recovery-policy contrast
398 gave $p = 0.375$ for the primary endpoint: seven of the eight cells whose final trees made the suite
399 uncollectable (five no-recovery, two repair-in-place) had been dropped as ungradable. Official grading
400 scores such a patch as failing every test, so the most contaminated states had been dropped from the
401 endpoint they bore on most. The grader, not the replay, now distinguishes an environment failure
402 from a patch that kills the suite, and the seven archived trees were re-graded without re-running any
403 agent.

404 C The public-scaffold arm

405 **The seam.** mini-swe-agent’s loop has one place that touches the working tree:
406 `DefaultAgent.execute_actions` calls `env.execute(action)`. We replace the environment

object and nothing else, so the scaffold keeps its own loop, prompts, litellm model layer, step and cost limits, and submission protocol. Our environment runs each bash command inside the benchmark’s pinned container, pulls the resulting tree back to the host the timeline records, and commits an observation when the tree changed. Two properties of the seam were established by measurement rather than assumed. The container’s git state is not advanced by the instrument, so the agent’s own `git status`, `git log`, and `git diff` show its uncommitted work and none of the instrument’s commits; the first paid pilot, run before this was true, spent eight of twenty commands discovering that its edits had been committed underneath it. The instrument is not invisible, only inert: the synthetic baseline commit it creates carries a fixed marker name, and 54 of the 80 runs saw that name in some command’s output, 8 of them through `git status` (departure 2). Whether seeing it changed behaviour we cannot settle here. The 54 runs that saw it resolved 43 and were bearing on 10; the 26 that did not resolved 24 and were bearing on 1 (one-sided Fisher $p = 0.069$). The obvious confound is that a run which edits more has both more to see and more to break, and it explains part of this: no run with two or fewer tree-changing observations was bearing at all, under either condition. Within the runs that edited more, though, the association does not vanish (10 of 27 against 1 of 11, $p = 0.088$), so we report it as an open question about the seam rather than a settled confound. And the routed command is wrapped so that a command ending in a here-document terminator or a comment parses as the scaffold’s own `bash -c` would parse it; before that fix, 39 of 40 GPT-5.6 cells hit a shell syntax error, because that model writes every script as a here-document.

Departures from the scaffold’s leaderboard configuration. (1) The measurement container has no network, while its own deployment uses Docker’s default bridge; the unchanged prompt tells the agent it may install tools, and here it cannot. (2) The container’s git history is a single synthetic commit, because the image’s upstream history can contain the fix the task asks for. (3) Resolve is graded on the final tree by the official grader rather than on the scaffold’s submitted patch. (4) Our replay and grading run outside its loop. (5) The two models are reached through the scaffold’s litellm layer, GPT-5.6 through its Responses-API model class, which the chat-completions class cannot serve (it returned three replies with no tool call and the scaffold exited on repeated format errors). (6) Prices come from our verified table rather than litellm’s, which carried a stale entry for one model. For these reasons we claim no equivalence to its published resolve rate.

Sampling grid. Per run, median tree-changing observations and mutated files, with the run and final observations excluded because neither is an opportunity to see a transition: public scaffold 3 and 3 (Claude), 2 and 2 (GPT-5.6); no recovery 2 and 2; repair-in-place 4 and 4; rollback 5 and 7; gate 4 and 4. Observations at which more than one file changed are 0 to 6% in every arm. The scaffold issues 14.8 and 9.1 bash commands per observation against the harness’s one tool call per observation, because most of its commands read rather than write.

The self-test classifier. For each episode we take the window in which the test was broken — the breaking command up to, but not including, the command that repaired it — and label three things over the scaffold’s own trajectory. *Tested*: a test runner ran and its output proves it ran, so a runner that died on an import does not count. *Covered*: one of those runs would have executed this test, decided on the runner’s positional arguments matched at a label boundary (a run of two sibling modules does not cover a test in a third), honouring `-k`, `-m`, `--deselect`, and explicit node ids. *Shown*: a covering run’s output names this test on a line that also carries a failure marker. Coverage and display are three-valued: a run that stopped at an earlier failure under `-x`, or whose output was truncated by a pipe, is *unknown* rather than negative, because calling it negative would assert that the agent ran the test and was not told, from evidence that cannot support it. The classifier’s labels were audited by hand against the raw trajectories of eight bearing cells from an earlier pilot of this arm (run before the two shell defects below were fixed, and therefore not among the runs reported here); the audit disagreed with an earlier version of the classifier on all three totals, and the version reported here reproduces the audit exactly.

The gate’s two contaminated Live trees. One run hit the \$4 cap before the gate saw its final tree; the other passed every check yet fails 13 grader tests, 7 in the file the hidden test patch rewrites.

Uses. The per-observation verdicts are the step-level ground truth that patch verifiers [24] and process reward models [25] require; we release the archived timelines for that purpose.

Further limitations. Event counts are lower bounds in both directions: a break that stops the suite from being collected opens no timeline event (Section 5.2), and no-recovery’s 19-of-19 persistence is confounded with its stopping rule, since that arm ends at the monitor’s first FAIL — a no-recovery

arm that keeps executing after a failed check is the experiment that would separate policy from truncation. The public-scaffold arm is one scaffold at one version under two model families; its budget differs in kind from our arms' (\$3 billed and 250 steps against a \$4 work cost), so the two harnesses are placed side by side and never tested against each other. The self-test measurement rests on 11 breaking edits, one of them coded unknown. Cross-stack comparisons confound model with adapter. The gated arms are post hoc and select tests from benchmark metadata. Live resolve is near floor, and under rollback most Live runs ship nothing, so its recovery contrast is uninformative on resolve.

Bounding the model-stack difference (Section 5.3). Exposure differs about twofold, not 140-fold: in opportunity units the Claude arm makes 147 observations to the GPT arm's 218, with per-run medians of 3 observations and 3 mutated files against 5 and 7. The zero is measured, not a detection failure: re-scoring that stack's earlier canary run reproduces its three events. And a second plain-rollback sweep reproduced the GPT side (13 of 36 graded resolved, 116 events, 6 of 37 bearing), so the bearing fraction is stable even though per-instance event counts are not.

The Claude rollback arm's contaminated tree. django-14238, a run whose tree never changed. Grading its unmodified base tree three times returns 38 of 39 previously-passing tests each time: one graded id never appears in the log, which upstream's rule scores as a failure. It is baseline rot of the kind Appendix A.3 describes, not a regression the timeline missed.

The two readings. The graded prediction excludes test files, as the official harness restores them anyway, while the replay reads the tree as it stands, so an agent edit to a test file is visible to the timeline and not to the grader.

Reading Table 1. *Baseline-dead* tests already fail in the unmodified image; *rot-aware* resolve counts a run resolved when every target test passes and no failure lies outside that set. On Live, **resolved** is of 40 while **empty** and **bearing** are of the 39 attempted (torch-tune-1806 exceeds the replay budget). **broken** → **left** omits any cell whose suite the grader could not collect, since such a cell would otherwise read as perfect capture; repair-in-place's row therefore covers six of its seven bearing runs and 83 of its 84 events.

Incidents. The pre-declared co-primary, per arm, in the order of Table 1: 5, 6, 11, 0, 7, 4, 7, 3, 9, 6, 6, 4.

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Answer: [Yes]

Justification: No models are trained. Section 4 gives the slices and how they were fixed, the harness and its three recovery policies (the two regression-gated variants are specified in Section 5.5), the public scaffold and its unmodified budgets, the exact model strings, the per-run cost cap, the outcome definitions, and the pre-declared analysis.

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Answer: [Yes]

Justification: The Claude rollback arm’s resolve carries an exact (Clopper–Pearson) 95% interval; paired comparisons use exact sign tests and exact McNemar tests, and the cross-stack comparison a one-sided Fisher exact test. The tests reported and the absence of multiplicity adjustment are declared (Section 5.5, Appendix B). Plain rollback was swept twice on Verified (Section 4) but the replicate’s numbers are not reported here, and intervals treat instances as independent although 16 of the 40 Verified instances come from django.

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Justification: Section 3 gives replay cost per observation and per sweep; Table 1 gives model spend per arm and substrate; Appendix B states the machine (one 12-vCPU, 22 GB x86 virtual machine running Docker, no GPUs) and the total model spend including pilots.

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